

Cosmetics



Chapter objectives

After completion of this chapter, You will be able to:

- ☐ Discuss the history of cosmetics
- ☐ Understand and mention the formulation and common ingredients of cosmetics ,
- ☐ Describe the different classes of cosmetics and their application

Introduction

- ❑ Cosmetics deals with those aspects of the skin related to beauty, concentrating on skin care, protecting the skin, and improving its appearance

Introduction...

- ❑ **Cosmetics:** are substances used to enhance or protect the appearance or odor of the human body (Related with beauty)
- ❑ Cosmetics include skin-care creams, lotions, powders, perfumes, lipsticks, fingernail and toenail polishes, eye and facial makeup, hair colors, hair sprays and gels, deodorants, baby products, bath oils, bubble baths, bath salts, butters and many other types of products.
- ❑ Their use is widespread, especially among women

Introduction...

❑ FDCA define cosmetics as :

“1) articles intended to be rubbed, poured, sprinkled, or sprayed on, introduced into, or otherwise applied to the human body or any part thereof for **cleansing**, **beautifying**, **promoting attractiveness**, or altering the **appearance** and
2) articles intended for use as a component of any such articles”.

Introduction...

- EFDA define cosmetics as :

“Cosmetic” means any substance or preparation intended to be placed in contact with the various external parts of the human body by means of rubbing, pouring, steaming, sprinkling spraying on or otherwise applied to the human body or any part thereof with a view exclusively or mainly for cleaning, perfuming, changing appearance and/or correcting body odors and/or protecting or keeping them in good condition and article intended for use as component of a cosmetic but such articles exclude laundry soap, articles intended for the diagnosis, treatment, mitigation or prevention of human diseases, and products intended to affect the anatomy or of a physiological process of a human;

Introduction...

- Classification in accordance with function
 - Those that improve appearance and beautify
 - impart a pleasant and attractive appearance
 - to camouflage less attractive areas
 - correct skin lesions
 - Those related to skin care
 - retain a smooth, soft and supple skin
 - Those related to skin protection
 - to shield the skin from the external effects of the sun, wind, cold

Introduction...

- Cosmetics
 - Doesn't require registration
 - No monograph
 - No pre-market approval
 - GMP Guidelines only, not regulation
 - Adverse events not required to be reported
 - not meant to affect the structure or function of

Introduction...

- THE GRAY AREA BETWEEN A DRUG AND A COSMETIC PRODUCT

Previously it was clear, now becoming blurred because of the below claims in marketed goods

- *“Accelerates the renewal of cells”*
- *“Builds up supportive tissue in the skin”*
- *“Repairs sun damage to the skin”*
- *“Repairs skin aging”*

- All the above changes function and/or structure

Introduction...

- Now the difference lies in the concentration of the active ingredient
 - low concentrations, -hydroxy acids function essentially as moisturizing agents; higher concentrations have any significant effect on the epidermis
- But still there is gray area between cosmetic treatments and dermatology, Cosmeceuticals
 - Medicated soap for acne, permanent make up

Introduction...

- ❑ Soap is specifically excluded from cosmetics in the FDCA (Food, Drug, and Cosmetics Act) and no cosmetic or drug regulations are applicable to soap.
- ❑ The devices used in applying the cosmetics as brush, comb, razor blades they are not included in cosmetics

History of cosmetics

- ❑ Spans at least 6000 years of human history, and almost every society on earth.
- ❑ Archeological evidence of cosmetics dates at least from ancient [Egypt](#) and [Greece](#)
- ❑ ancient Egypt used castor oil as a protective balm
- ❑ [skin creams](#) made of [beeswax](#), [olive oil](#), and [rosewater](#) described by the Romans

History of cosmetics...

- ❑ The Palestine, Jezebel painted her eyes—approximately 840 BC recorded in the Old Testament—2 Kings 9:30
- ❑ The Biblical book of [Esther](#) describes various beauty treatments as well.
- ❑ Cosmetic use was frowned upon at some points in history.
 - ✓ in the [1800s](#), make-up was used primarily by [prostitutes](#), and [Queen Victoria](#) publicly declared makeup improper, and acceptable only for use by [actors](#)

- ❑ Although use of cosmetics have existed since the origin of civilization, only in the 20th century has great progress been made in the diversification of products and functions and in the safety and protection of the consumer.
- ❑ Before 1938, cosmetics were not regulated as drugs, and cosmetology could often be considered as a way to sell dreams rather than objective efficacy; safety for consumers was also sometimes precarious.
- ❑ FDA through the Federal Food, Drug, and Cosmetic Act, regulated cosmetics that were required to be safe for the consumer

Principles in the Preparation of Medical and Cosmetic Product

- All Preparations for external use (cosmetics and medical products) contain 3 components
 - an active ingredient,
 - a base (or vehicle), and
 - an additional/auxiliary substance
- No active ingredient meant to be applied to the skin in its pure chemical form
 - It is combined with a base and with other substances for effective preparation and Skin Penetration

Principles...

- an active ingredient
 - Main component of the cosmetics whose action produces the main effect
 - Inhibit aging of the skin (e.g., retinoic acid or an alpha-hydroxy acid)
 - lightens dark lesions on the skin (e.g., hydroquinone).

Principles...

- Base (Vehicle)
 - Carries the API
 - must ensure that the active ingredient remains chemically stable, can penetrate the skin, and can be released effectively within the skin
 - The three elemental bases are:
 - fatty bases,
 - aqueous (water) solutions and
 - Powders
 - Additional effect, increasing the moisture level, soothing, or cooling

Principles...

- Additional/Auxiliary Substances
 - Medicinal or cosmetic preparations usually also contain other substances.
 - Common additives are:
 - fragrances and perfumes,
 - dyes, and
 - preservatives

Principles...

- Fatty Bases
 - Good for skin Penetration
 - Moisturizes by creating film
 - derived from animal sources, plant sources, or mineral sources
 - Can be solid (Wax), liquid (Oil) or semisolid (Fat)

Principles...

- Animal-Derived Fatty Bases
 - Lanolin
 - Derived from sheep's wool, from sheep's sebaceous glands, commonly used for moisturizers
 - Non-irritating as it is similar with human sebum
 - Wool Alcohols
 - Contain more alcoholic oily compounds than does lanolin. Lanolin contains more water
 - Spermaceti
 - From whales, commercial gimmick (No special advantage)

Principles...

- Plant-Derived Fatty Bases
 - olive oil, sesame oil, peanut oil, corn oil, sunflower oil, soy oil, and cocoa butter
 - Efficacy is same as animal derived
 - ratio of polyunsaturated to saturated oils in their composition is irrelevant

Principles...

- Fatty Bases Derived from Minerals
 - paraffin oils, which are products of the refining of crude oil
 - liquid paraffin (liquid petroleum),
 - semisolid paraffin, white soft paraffin (petroleum jelly, semisolid petrolatum), or
 - solid paraffin (wax)
 - Paraffin is an efficient occlusive
 - inert, it very rarely causes skin irritation
 - In general, different fatty bases can be mixed together to achieve the desired properties.

Principles...

- BASES IN COSMETIC AND MEDICAL PRODUCTS: AQUEOUS SOLUTIONS
 - The most common liquid used in cosmetic preparations is water
 - Water has additional effects on the cooling and drying of the skin
 - wetting the skin frequently with water actually has a drying effect!
 - when we want to dry out inflamed, weeping areas of skin, we repeatedly wet those areas

Principles...

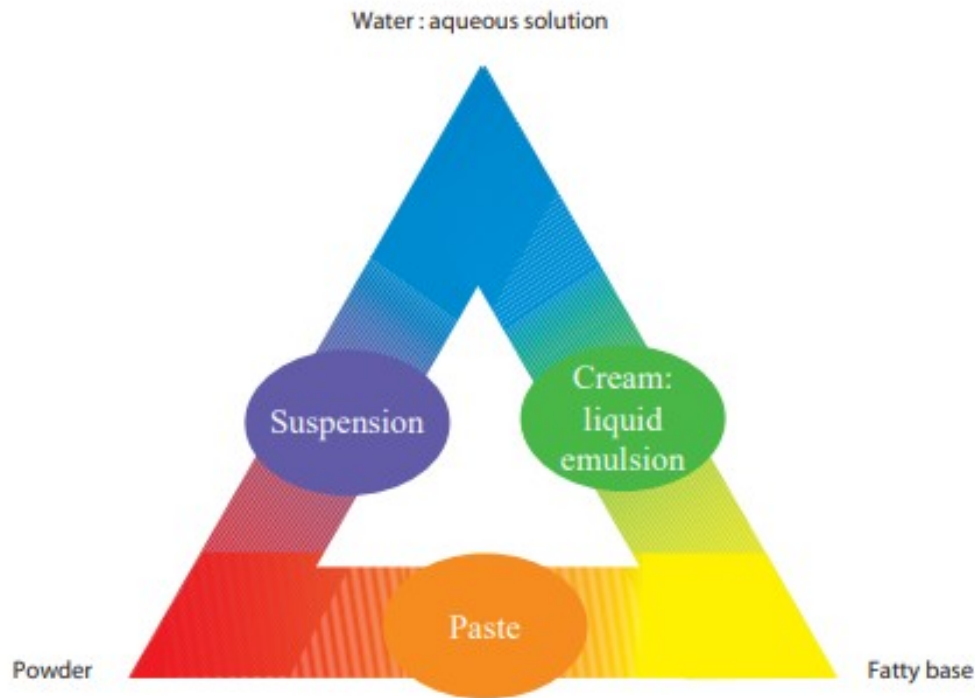
- BASES IN COSMETIC AND MEDICAL PRODUCTS: POWDERS
 - Fine powders that do not contain coarse particles
 - to prevent friction and to absorb excess moisture
 - usually used in skin creases (such as the groin)
 - Cover and conceal certain areas of the skin.

Principles...

- Substances Commonly Used in Powders are
 - Zinc Oxide (Cover and protect),
 - Titanium Dioxide (UV protect),
 - magnesium polysilicate (talc) (Inert, for protection),
 - Calamine, mixture of zinc oxide with a small amount of iron oxide (Soothing effect and decrease itching)
- In general, it is not advisable to use pure powders on babies

Principles...

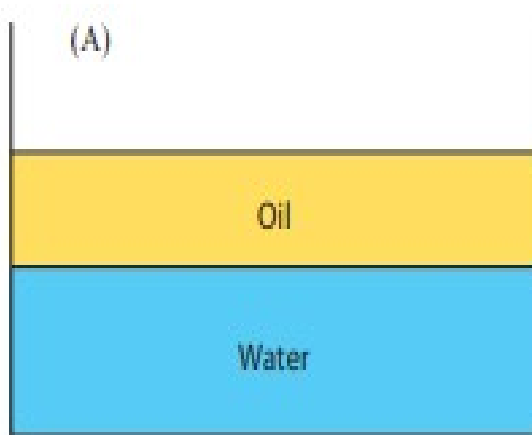
- COMBINATIONS OF BASES



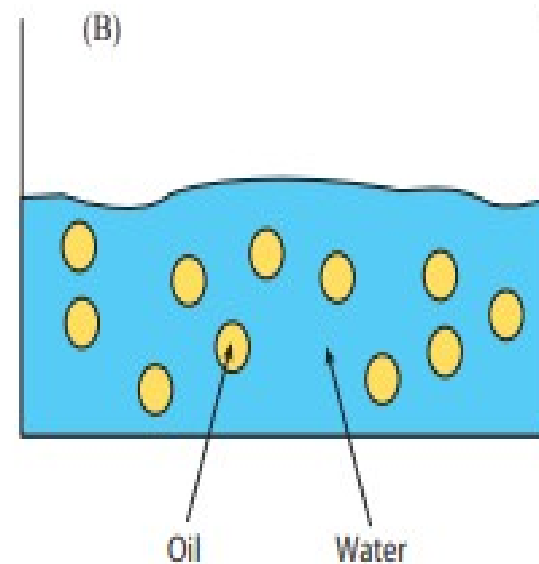
Triangle illustrating the various possibilities in combining bases.

Principles...

- COMBINING A FATTY BASE WITH WATER
 - Lotions, Cream and Emulsion



(A) Vessel containing oil and water.



(B) After mixing the oil and water.

Principles...

- There are many types of cream in cosmetics, each used for its specific purpose, and advertised under different names:
 - vanishing creams
 - night creams
 - cleansing creams
 - moisturizing creams
 - foundation creams
 - cold creams
 - eye creams.

Principles...

- Vanishing Creams
 - Relatively high water content, easy to wash off
 - has a cooling effect.
 - have some oil content, moisturizing effect
 - used as daytime creams
 - sunscreens , antibiotics, retinoic acid are added
 - are transparent, and invisible once they have been applied.

Principles...

- Night Creams
 - Higher oil content, greasier more occlusive than vanishing creams
 - intended for use on dry skin, no cooling effect
 - also known as “nourishing” creams
 - Hence, “nourishing” creams are applied at night, before going to bed. Because they are greasy, they tend to stay on the skin longer.

Principles

- Cleansing Creams
 - Composed of oil, water, and cleansing agent.
- Moisturizing Creams
 - designed to increase the skin moisture content, are based on occlusives, which produce an impermeable barrier on the skin surface, and on humectants, which absorb water.

Principles...

- Foundation Creams Foundation
 - moisturizing creams
 - usually contain coloring agents and sunscreen
 - provide a smooth, even color to the face and are used to conceal skin blemishes
 - These creams are available in a range of shades, so that every woman can find the appropriate color for her skin

Principles...

- Cold Creams
 - Have a cooling effect
 - The cooling occurs because these creams are “pseudo” emulsions, rather than true emulsions
 - A cold cream is a simple mixture of oil and water with out emulsifier, and so is not a stable product
 - Hence, when applied to the skin, the water separates from the oily component, and quickly evaporates and creating a cooling effect

Principles...

- Eye Creams
 - Applied to the delicate skin around the eyes are mainly hypoallergenic preparations
 - they do not contain components such as certain perfumes and/or certain
 - Nevertheless, there is no doubt that even eye creams may result in allergic reactions in some people, albeit very rarely

Principles...

- COMBINING A POWDER WITH WATER:
SUSPENSIONS
 - Solution
 - Tincture
 - Suspension
- Once the water has evaporated, a layer of powder remains on the skin , so as to spread the particles of powder evenly throughout the liquid

Principles...

- COMBINING A POWDER WITH A FATTY BASE
 - PASTES
 - The fatty base is usually petrolatum
 - less greasy than regular ointment, but it still has occlusive and protective properties
 - Because a paste contains powder, it has the ability, as opposed to an ointment, to absorb liquids to a certain extent
 - Commonly used for diaper rash
 - Used for medical than cosmetics

Principles ...

- ORGANIC COSMETIC PREPARATIONS
 - Is not supposed to contain chemicals and artificial additives such as preservatives, synthetic coloring agents, or artificial fragrances
 - perceived by the public as healthy and safe
 - In practice, the manufacturing of a 100% organic product cannot always be implemented
 - Short shelf life

- ❑ The target substance in cosmetic products may not be easily applied on the human body
 - ✓ may be dissolved into or
 - ✓ mixed with different kinds of chemical additives
- ❑ Excipients may be used to bulk up cosmetical formulations with very potent target substances, to allow for convenient and accurate dosage

Formulation and common ingredients...

- ❑ Once cosmetic products have been formulated, it may not stay in active form for long
 - ✓ will denature,
 - ✓ fall out of solution,
 - ✓ or stick to the sides of the container
- ❑ To stabilize the product, some **excipients** are added, ensuring that the target ingredient stays "active", and, just as importantly, stable for a sufficiently long period of time

Formulation and common ingredients...

- ❑ All ingredients in cosmetics, as well as their [chemical decomposition](#) products are to be identified and guaranteed to be safe.
- ❑ excipients are only used when absolutely necessary and in the smallest amounts possible.

Formulation and common ingredients...

Fillers/Diluents/vehicle/Bases

- ❑ Fillers fill out the amount of the product, making it practical to produce and convenient for the consumer to use

- ❑ Provide proper volume for consumer handling
 - a varied group of oily or water soluble polymers, as discussed before

Formulation and common ingredients...

Color additives

- ❑ added to improve appearance of a formulation
- ❑ vital to any product that is intended to alter the color of any part of the body
- ❑ Color materials mixed in with cosmetics can be broadly classified in to organic synthetic coloring agents , natural colors, inorganic pigments and polymer powders

Formulation and common ingredients...

Lubricants

- prevent ingredients from clumping together and from sticking on the processing machines

An antioxidant

- preventing the oxidation of other additives
- Oxidation reactions can produce free radicals, which start chain reactions
- Antioxidants terminate these chain reactions by removing free radical intermediates and inhibit other oxidation reactions by being oxidized themselves

eg. thiols or polyphenols.

Formulation and common ingredients...

Fragrances

- Give the product a fresh, pleasing odor and cover up the smell of other ingredients
- Fragrances in skin-care products are the most likely cause of skin irritations or contact allergies
 - eg cinnamic alcohol, hydroxycitronella and isoeugenol

Formulation and common ingredients...

Preservatives

- Any product that includes water and oil must contain one or more preservatives
- may cause skin reactions
- eg diazolidinyl urea, imidazolidinyl urea, quaternium 15, and the parabens, benzoic acid, formaldehyde.

An emulsifier

- a type of surfactant typically used to keep emulsion well dispersed.

Formulation and common ingredients...

Humectants

- Maintaining a young-looking is closely connected with moisture content
- The keratin layer of the skin contain natural moisturizing factors (NMF) with hydrophilic moisture absorbing compounds
 - NMF play important role in skin moisturizing
 - Sodium salt of pyrrolidonecarboxylate is the most important factor in NMF
- Cosmetics should be able to mimics this natural moisture retention mechanism

Formulation and common ingredients...

- Humectants are water soluble materials with high water absorption and they are a very important component in the aqueous phase of cosmetics
- They absorb water from the air and hold the moisture in the skin
 - useful in softening thickened or scaly skin
- Humectants need very high humidity levels to be effective
 - e.g. polyhydric alcohols like, glycerin, propylene glycol, sorbitol
 - main components of NMF , propylidonecarboxylate and lactates

Formulation and common ingredients...

Main requirements of humectants are

- Must have appropriate water absorption ability
- Must maintain water absorption ability
- Water absorption must not be influenced by changing environmental conditions (temperature ,humidity)
- Water absorption ability must maintain moisture in the skin
- Must have lowest possible volatility
- Must have good miscibility with other constituents
- Must be safe
- Must be colorless, odorless and tasteless
- Must have densities matching the system they are used in

Formulation and common ingredients...

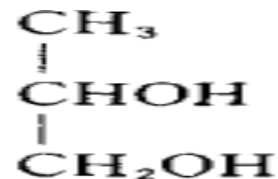
Glycerine

- Obtained as a byproduct in manufacturing of soap from plant and animal fats and oils
- When dehydrated and deodorized, it is colorless, odorless liquid



Propylene glycol

- It looks and feels much like glycerin and is colorless and odorless
- It has a better feeling on use due to its lower viscosity than glycerin



Formulation and common ingredients...

sorbitol

- It is white odorless solid and obtained by reduction of glucose
- it has lower hygroscopic property
- It has a humectant effect at low humidity and used in creams, lotions, toothpaste

Sodium lactate

- humectants occurring in NMF along with pyrrolidonecarboxylate
- have a higher water absorption ability

Formulation and common ingredients...

Sodium 2-pyrrolidone-5-carboxylate

- Excellent hygroscopic and humectant effect

sodium hyaluronate

- It is sodium salt of Hyaluronic acid and a white water-soluble powder
- Properties like viscosity and humectance vary with the Mwt
- it is almost unaffected by environmental humidity

Formulation and common ingredients...

Emollients

- These ingredients fill in the spaces between the cells in the skin
 - ✓ thus soothing and lubricating rough skin
- eg butyl stearate, glycerin, lanolin, mineral oil, petrolatum

Formulation and common ingredients...

Ultraviolet Filter

- UVR reaching the Earth's surface can be divided into UVB (290–320 nm) and UVA (320–400 nm)
- UVA can be further subdivided into UVA I (340–400 nm) or far UVA and UVA II (320–340 nm) or near UVA
- UV filters have been traditionally divided into **chemical absorber** and **physical blockers** on the basis of their mechanism of action.
- **Chemical absorber** are generally aromatic compounds conjugated with a carbonyl group

Formulation and common ingredients...

- These chemicals absorb high-intensity UV rays with excitation to a higher-energy state. The energy lost results in conversion of the remaining energy into longer lower-energy wavelengths with return to ground state.
- The main UV absorbent used in cosmetics
 - benzophenone derivatives (tetrahydroxybenzophenone)
 - para-amino benzoic acid derivatives (para-amino benzoic acid(PABA)),
 - paramethoxy cinnamic acid derivatives (ethyl-paramethoxycinnamate)
 - salicylic acid derivatives (octyl salicylate)

Formulation and common ingredients...

- **Physical blockers** reflect or scatter UVR.
- Sometimes referred to as “non-chemical” sunscreens
- Newer micro sized forms of physical blockers may also function in part by absorption
- they may be more appropriately designated as inorganic particulate sunscreen ingredients

E.g Titanium dioxide and Zinc Oxide

Formulation ingredients of cosmetics ...

➤ The important requirements of UV filters used in cosmetics are

- Non-toxic
- High UV absorbance or reflection or scattering over a wide range
- No break down due to UV light and heat
- Good compatibility with cosmetics base materials

Classification of cosmetics

1. Depending up on the physical form cosmetics can be classified as

- Emulsions eg: cold cream; vanishing cream
- Powders eg: face powder: talcum powder, tooth powder.
- Sticks eg: Lipsticks, deodorant sticks
- Oils eg: hair oils.
- Jellies eg: hair jelly, hand jelly
- Paste eg: Tooth past, deodorant paste
- Solution eg: After shave solution
- Aerosols eg: After shave spray, hair sets, perfumes
- Suspensions: eg cosmetics stockings

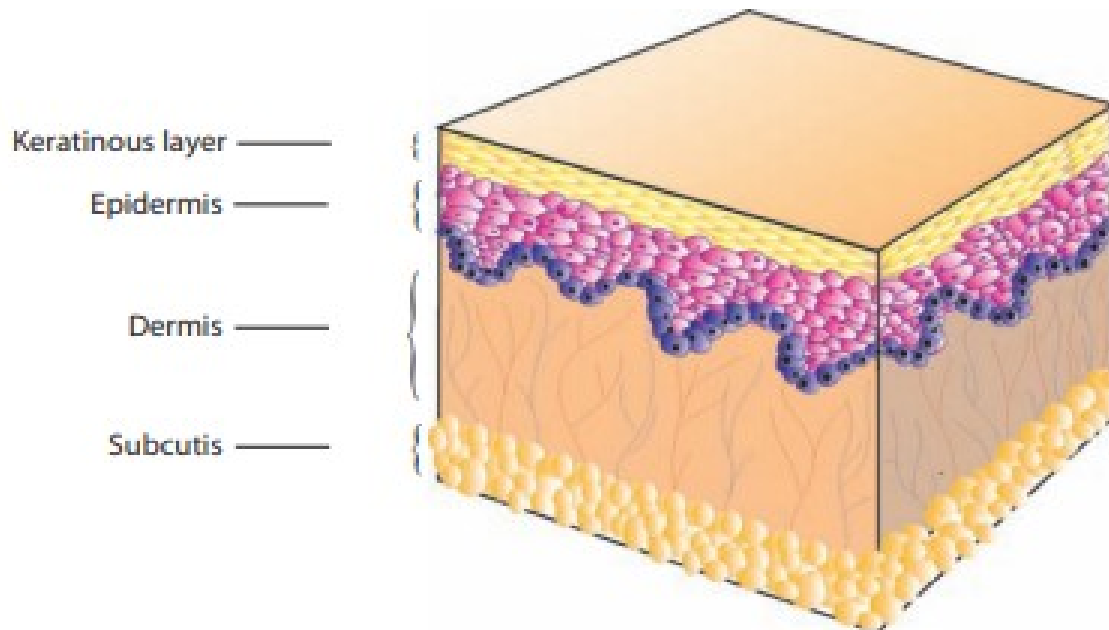
Classification of cosmetics...

2. Can be classified based on widely used body parts as:

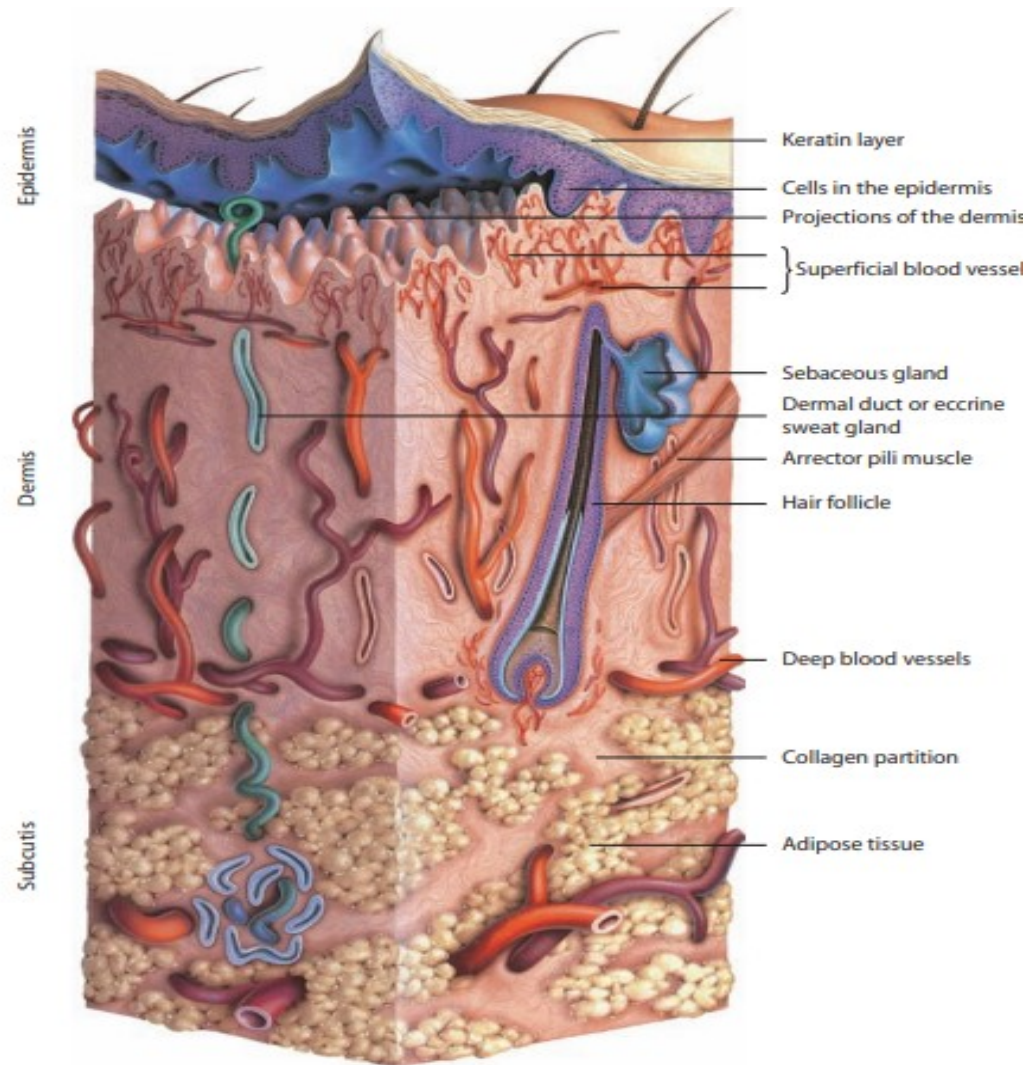
- Hair cosmetics
- Face cosmetics
- Eye make ups
- Lip decorators and carers
- Nail cosmetics
- Skin cosmetics
- Oral cosmetics

Skin Cosmetics

- structure and function of the skin



Skin Cosmetics...

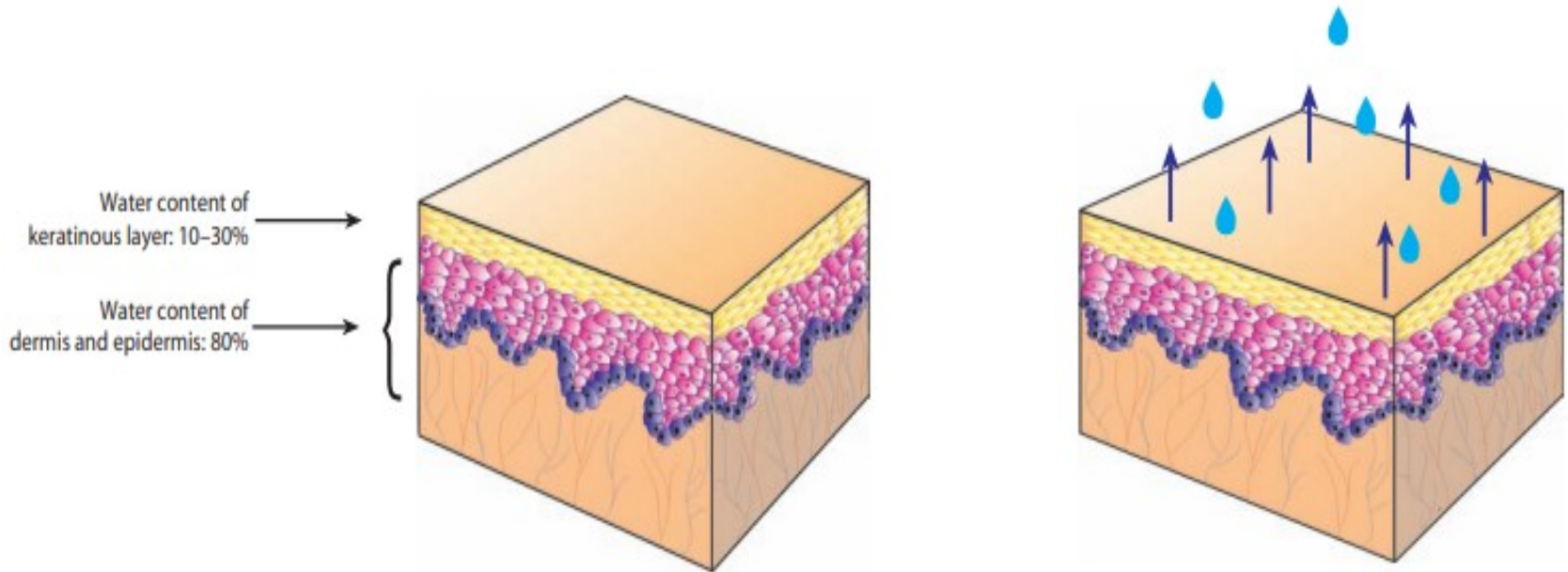


Transverse section through the skin

Skin Cosmetics...

- FUNCTIONS OF THE SKIN
 - protective layer,
 - transmits sensations,
 - helps to regulate the body's temperature,
 - produces vitamin D, and
 - plays a role in social interactions.

Skin Moisture and Moisturizers



When the water content of the skin is normal, it appears soft, smooth, supple, and glowing. The skin is slightly filled out with water, which tends to smoothen the fine wrinkles.

In normal skin, there is a continuous movement of water from the deep layers to the superficial layers. Eventually, the water evaporates from the surface.

Skin Moisture and...

- Dry Skin
 - External Causes
 - exposure to dry environments and wind
 - Heater, winter time
 - washing, and
 - Frequent washing removes oily layer
 - exposure to certain substances
 - Exposure to detergent
 - The Skin's Ability to Retain Moisture
 - Age
 - skin disorders like atopic dermatitis

Skin Moisture and...

- CHARACTERISTICS OF DRY SKIN
 - dry, fissured, and rough
 - Fine lines are more apparent
 - feeling of dryness, accompanied by itching
 - superficial layers do not peel off easily and remain attached
- prone to skin infections, both bacterial, and fungal.
- The common dermatological term for extremely dry skin is xerosis

Skin Moisture and...

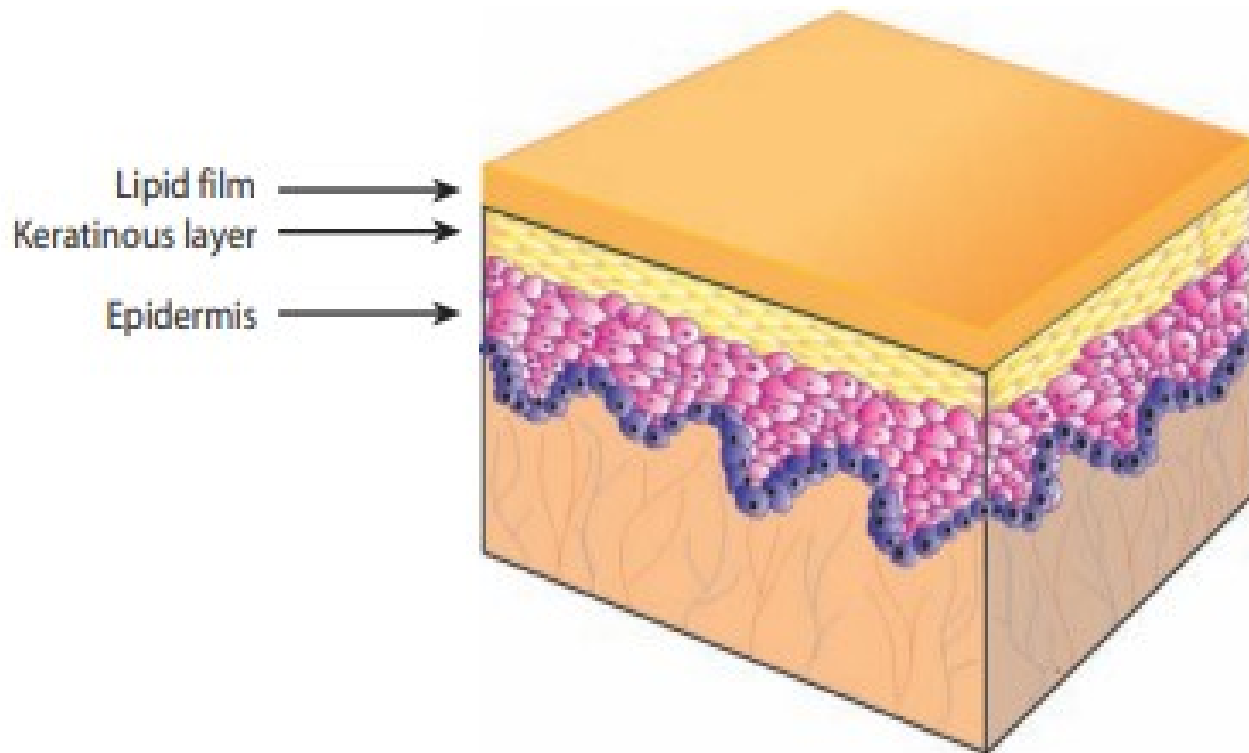
- NATURAL FACTORS THAT PREVENT SKIN DRYNES
 - An oily layer on the skin, Lipid film
 - decreases water evaporation, occlusive layer above the keratinous layer
 - a combination of oily products and various lipid degradation products
 - Natural moisturizing factor
 - urea, lactic acid, glycolic acid, phospholipids, malic acid, pyruvic acid, and salts of pyrrolidone carboxylic acid

Skin Moisture and...

- **MOISTURIZERS: OCCLUSIVES AND HUMECTANTS**
 - Occlusives
 - oily layer on the skin, enriching the skin's natural lipid film, which prevents water evaporation
 - The keratinous layer dampens, becoming more fully saturated with water.
 - more effective if applied directly after washing, trapping a layer of water beneath them

Skin Moisture and...

- Occlusive



Skin Moisture and...

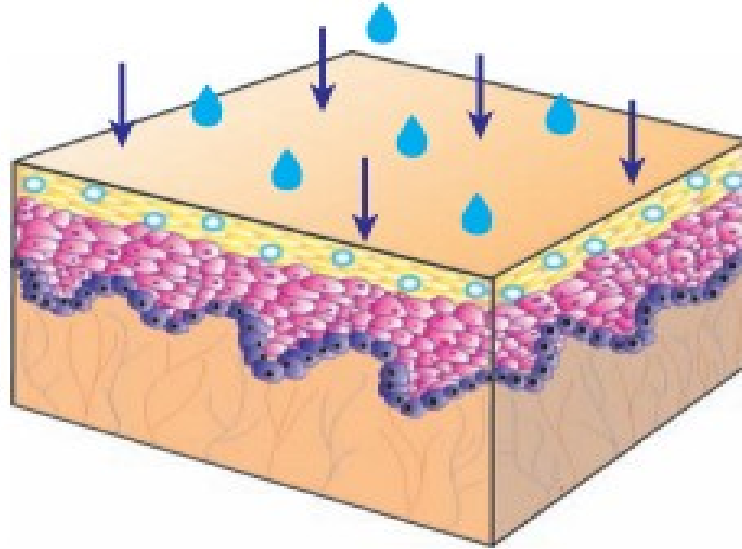
– Occlusive substances

- Mineral-derived fatty, paraffin, Cetomacrogol
 - highly effective, but they are sticky and greasy, for very dry skin
- animal fat, Lanolin
- Vegetable oils such as olive oil, oat oil, peanut oil, sesame seed oil
 - Vegetable oils are less occlusive than animal-derived

Skin Moisture and...

– Humectants

- Works by absorbing water
- penetrate the keratinous layer and increase its water content



Skin Moisture and...

- In a relatively arid environment, causing increased dryness, specially macro molecules. in a humid environment, it is efficient.
- Substances as humectant are
 - small molecules like glycerine (glycerol) sorbitol propylene glycol
 - Macromolecules like glycosaminoglycans (such as hyaluronic acid), elastin, collagen, and other proteins.
 - natural moisturizing factor like sodium salts of pyrrolidone carboxylic acid, urea (in 10% to 20% concentrations), lactic acid, and phospholipids

Skin Moisture and...

- Therefore, an efficient moisturizer suited to cold, dry weather, should contain a combination of occlusives and humectants
- Most moisturizers contain occlusive, oily products along with humectants

Skin Moisture and...

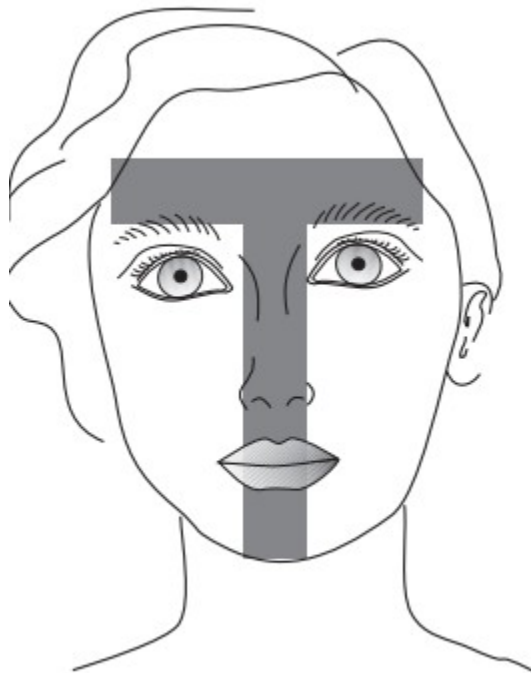
- HOW TO SELECT A MOISTURIZER
 - 100s of moisturizers in market which varies in water content and lipid components, with DF of Liquid emulsions, Ointments Creams
 - The foremost factor in selecting a moisturizer is the skin's lipid content, Skin types
 - Dry Skin
 - The pores are hardly noticeable
 - scaly and fissured
 - Lighter toned skin

Skin Moisture and...

- Oily Skin
 - glossy, especially on the forehead, nose, and chin
 - Large pores are apparent
 - tend to suffer from acne as adolescents
- Normal Skin
 - e between dry and oily skin
 - neither glossy nor oily to the touch
 - appears smooth and well moisturized
 - The pores are not large

Skin Moisture and...

- Combination Skin
 - identical to the normal skin type
 - Oily T-zone



The T-zone: forehead, nasal bridge, nose, and center of the chin.

Skin Moisture and...

- Which Moisturizers Should Be Used for which Types of Skin?
 - Dry facial skin, only humectants will not be enough. Oily moisturizers containing occlusive components are required. If the skin is fairly “normal” and not dry, one should use a preparation that has both occlusives and humectants
 - oily skin or a tendency toward oily skin does not need moisturizers, except in windy and hot condition

Skin Moisture and...

- normal or near normal, less of the oily, occlusive substances. In this case, preparations containing humectants are recommended
- Combination, avoid moisturizing preparations in the T-zone, On the rest of the face, one should use nonoily products (containing humectants).
- Age and special weather condition may require moisturizer for normal skins as well

Skin Moisture and...

- GUIDELINES FOR THE USE OF MOISTURIZERS
 - Dry skin, should avoid washing with soap, avoid exposure to harsh environment,
 - Apply after skin cleansing, when damp
 - gentle application,
 - Frequent application for dry skin
 - Apply the moisturizer 20 to 30 minutes prior to exposure to a cold, dry, or windy environment. Under these conditions, oily moisturizers may be preferable.

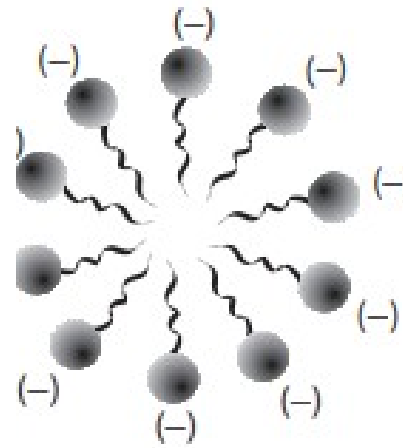
Skin Cleansing

- What is the dirt that has to be removed?
 - dust
 - soot (from the air)
 - Sweat
 - breakdown products of sebum
 - Residues of cosmetics and makeup previously applied to the skin
 - other substances carried in the air which vary depending on the geographical location and immediate environment.
- Water alone is not effective enough to cleanse the skin

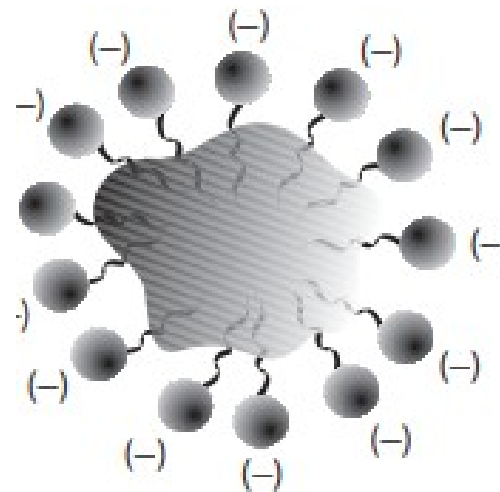
Skin Cleansing...

- SOAP AND ITS MODE OF ACTION
 - consist of salts of various fatty acids
 - Stearic acid
 - Palmitic acid
 - Oleic acid
 - Myristic acid
 - Lauric acid
 - regular, classic soap, known as hard soap or toilet soap, comprises the sodium salts of fatty acids

**These fatty acids are
derived from either
animal or vegetable
sources**



A soap micelle.



A micelle coating a fat globule.

Skin Cleansing...

- POSSIBLE DISADVANTAGES OF REGULAR SOAP
 - “sticky,” not readily soluble salts are formed because of Ca and Mg in Water, lead to skin irritation
 - high pH (9-10), also cause irritation
 - normal skin pH (which is between 4 and 6.5)
 - In some people, abrupt changes in pH can cause significant skin irritation
 - Therefore, to adapt the pH of cleansing agents and other cosmetic preparations to that of normal skin is the current trend

Skin Cleansing...

- SYNTHETIC SOAPS (“SOAPLESS SOAP”)
 - To Overcome the disadvantage of regular soap
 - Derived from by-products of crude oil refining
 - Surfactants, or surface-active agents make up the major components of soaps and shampoos
 - Same mechanism as regular soaps
 - All cleansing agents made of surface-active agents are called artificial soaps, soapless soaps, nonsoaps, or synthetic soaps

Skin Cleansing...

Surfactants

There are four groups of surfactants such as:

- anionics,
- cationics,
- nonionics, and
- amphoterics.



The nature of each group is determined by its chemical charge. Each surfactant group has different chemical properties that affect the way it cleans.

Skin Cleansing...

- Detergents are soapless soap, though people associate it with strong detergent for dishes and other goods
- Possible Advantage of Synthetic Soaps is PH adjustment, otherwise it is more irritating than the regular soaps
- “INTERMEDIATE” GROUP OF SOAPS
 - combination between regular and synthetic soaps. pH lies somewhere between the two types

Skin Cleansing...

- WHAT DO SOAPS CONTAIN APART FROM THE ACTIVE INGREDIENTS (Surfactants or fatty acid salts)?
 - Moisturizers
 - Preservatives
 - Coloring agents
 - Fragrances and perfumes
 - Antibacterial substances
 - Substances that alter the pH, and other ingredients.

Skin Cleansing...

- Creams and Liquid Emulsions for Facial Cleansing
 - Water rinses off mixture of soap and oil from the skin in regular soap
 - cleansing creams and emulsions already contain water
 - Applied to the skin with the fingertips and left on the skin briefly, remove with a tissue or a wet facecloth, or by rinsing off with water (preferred!), the fatty layer with the dissolved dirt is removed with them.

Skin Cleansing...

- CREAMS/EMULSIONS Vs SOAP
 - CREAMS/EMULSIONS contain high fat content
 - Better for sebum and makeup removal
 - Recommended for dry skin, not for oily skin as thin layer is remained in the skin

Skin Cleansing...

- **ABRASIVE CLEANSERS**
 - the abrasive effect is achieved by fine granules
 - supposed to remove the keratin layer by mechanical means
 - Removing the outermost layers of keratin may help produce a uniform, smooth surface on the skin and improve its appearance.
 - Not recommended for healthy skin,
 - Possibility of damage to the epidermis if this abrasive cleaning is carried out too vigorously and roughly
 - Should be gentle rub and used rarely

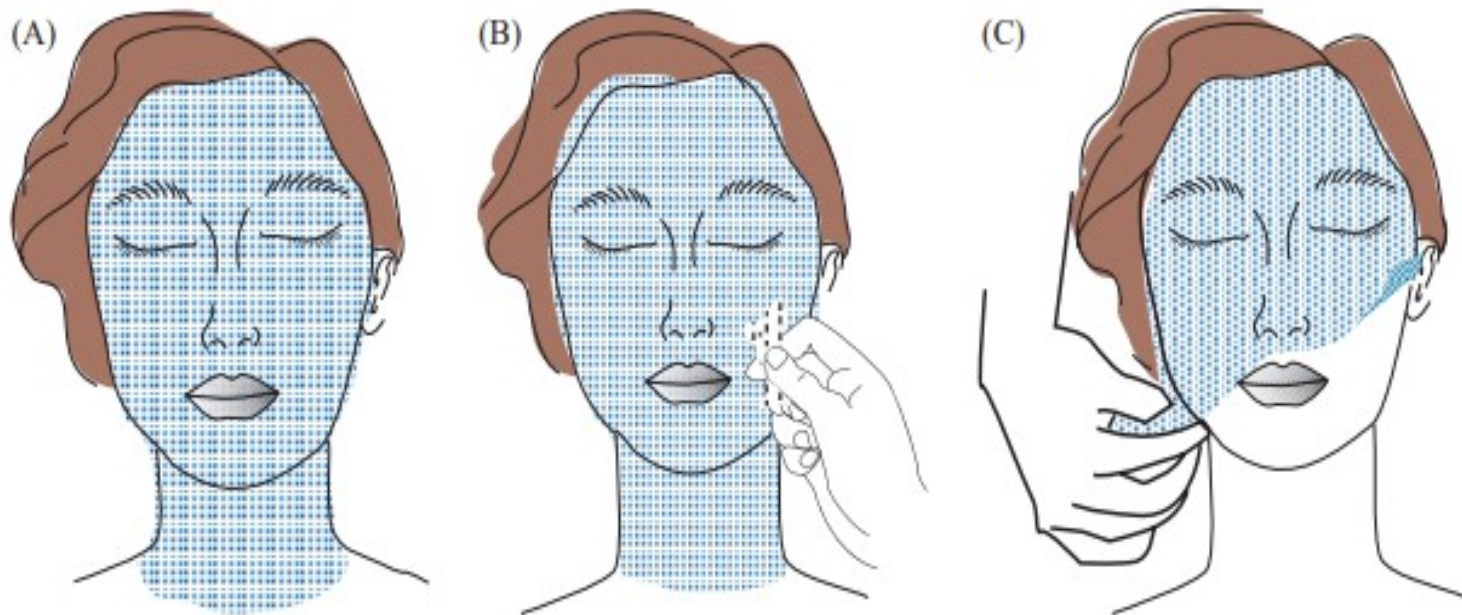
Skin Cleansing...

- Facial Cleansing Masks
 - applied as a relatively thick layer and then removed some time later (15-30 min), with same mechanism as soap
 - Functions
 - Effective cleansing of the skin
 - Moisturizing the skin
 - Treatment of acne
 - Improvement in the overall feeling of well-being

Skin Cleansing...

- MASKS THAT RINSE OFF AND MASKS THAT PEEL OFF
 - Masks That Are Rinsed off
 - Rinsed off by lukewarm or warm water
 - consist of absorbent masks, which are based on insoluble powders, natural clay and mud, or gel masks
 - contain ingredients such as tragacanth, zinc oxide, titanium dioxide, kaolin, calamine
 - Masks That Are Peeled off
 - made of rubbery substances, such as polyvinyl alcohol or rubber based substances such as latex or other natural rubber compounds
 - recommended for women with relatively dry facial skin

Skin Cleansing...



(A) A piece of gauze soaked in water is placed over the face. (B) The mask is applied to the face on top of the gauze. (C) After the required treatment time, the gauze is lifted and rolled up off the face, taking the mask with it.

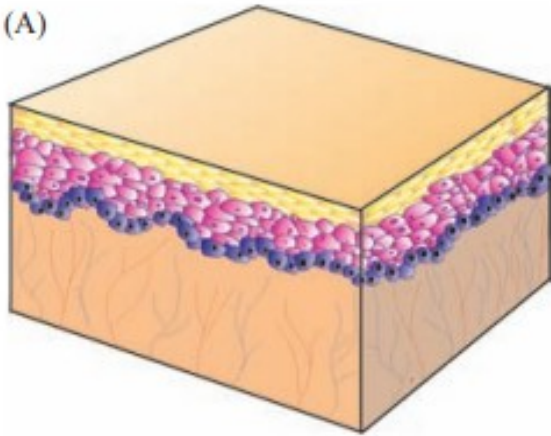
Skin Cleansing...

- POSSIBLE UNDESIRABLE EFFECTS FROM FACIAL CLEANSING MASKS
 - skin irritation
 - Skin infection
 - These complications are more likely to occur from the use of masks of dubious origin. The risks of such problems are much fewer when using masks from a reputable cosmetics manufacturer. In general, before using any mask, one should establish that the client is not allergic to any of its ingredients.

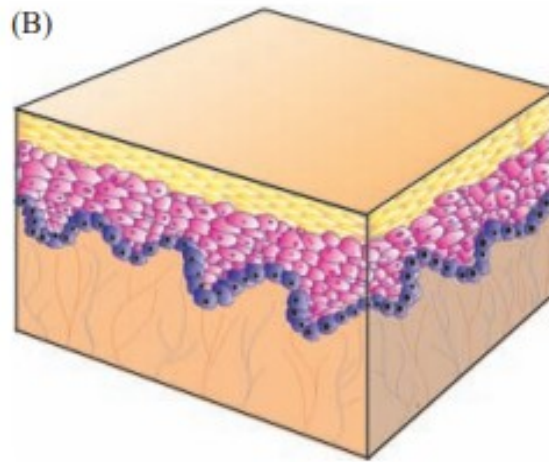
Skin Aging and Its Management

- Skin Aging and Its Management
 - Young age:- soft, smooth, and supple skin
 - Old Age:- wrinkles, age spots, skin loosening of the skin.
 - chronological aging
 - environmental factors
- SKIN AND AGE: CHRONOLOGICAL AGING
 - degeneration of elastin fibers
 - degeneration of collagen fibers, and
 - thinning of the skin.

(A)



(B)



Flattening of the attachment between epidermis and dermis in older skin (A), compared with the wavy attachment in younger skin (B).



Extremely thin atrophic skin.

Skin Aging and...

- Additional Changes that Appear with Age
 - With increasing age, significant changes occur in the skin that affect:
 - moisture content
 - rate and location of hair growth
 - pigmentation, and
 - the size of the sebaceous gland

Skin Aging and...

Chronological Aging

Thin, atrophic skin

Degeneration of collagen and elastin fibers

Possible development of skin tumors

Lighter skin due to decline in melanin production

Additional features such as:

- drier skin,
- changes in hair growth, and
- enlargement of sebaceous glands.

Photoaging

An irregular pattern of thicker skin; an increase in the number of epidermal cells

Accelerated degeneration of collagen and elastin fibers

Possible development of skin tumors, which are typical of photoaging

Uneven pigmentation: appearance of “age spots” (solar lentigines)

- Telangiectases (dilated blood vessels in the skin)

Differences between chronological aging and photoaging.

Skin Aging and...

- HOW TO CONTROL SKIN AGING
 - Avoid excess sun exposure
 - Don't smoke
 - Prevent unnecessary stretching of the skin
 - Change facial expressive habits, if necessary
 - Use certain topical products
 - Lead a healthier emotional and physical lifestyle.

Skin Aging and...

- Use of Certain Topical Products
 - Skin aging, was previously considered to be irreversible
 - Few years back, new products that can affect the aging process have been developed, revolutionizing cosmetic dermatology.
 - retinoic acid, and
 - Alpha-hydroxy acids
 - No clear data as to the effectiveness of other products, such as herbal extracts and preparations containing topical vitamins
 - Moisturizers improve skin health, but not antiaging

Skin Aging and...

Anti-aging cream

- moisturiser based skin care products marketed with the promise of making the consumer look younger by reducing visible wrinkles, expression lines, blemishes, pigmentation changes, discolorations of the skin.
- Despite great demand, many such products and treatments have not been proven to give lasting or major positive effects.
- A decrease in wrinkle depth of 10% is typical.



Skin Aging and...

- In Conventional moisturising ingredients anti-aging creams usually contain anti-aging ingredients
 - [Retinol](#) (in the form of [retinyl palmitate](#)): to reduce fine lines and pores.
 - [Epidermal Growth Factor](#): to stimulate cell renewal and [Collagen](#) production in the skin and strengthen elasticity and structure.
 - reduce fine lines, wrinkles and sagging.
 - [Alpha hydroxy acids](#) (AHAs) help to dissolve the intracellular "glue" that holds the dead cells together on the skin and enhances the [exfoliation](#) of the dead cells.
 - This exposes newer skin cells and can help improve appearance
 - Antioxidant

Hair cosmetics

Shampoos

- are based on soap or synthetic detergents
- used to cleanse, remove excess dirt, oil, and unwanted build-ups from the hair and scalp and leave the hair lustrous (Generally, removing sebum and oil)
- There a great variety of shampoos which have value in the form of conditioning, luster enhancing and styling capabilities, for people with dandruff, hair fall, colour-treated hair, etc.. And also for men, women, young

Ingredients used In shampoo include

- Opacifiers: used to make the shampoo opaque
 - Glycol and glyceryl stearate , cetyl and stearyl alcohol and stearic amides

Hair cosmetics...

- Solublizing agent: used to solubilize poorly soluble substances so as to get a clear shampoo
 - Ethyl alcohol, glycerol, propylene glycol
- Thickening agents: increase the viscosity of the shampoo and provide the desired consistence for the preparation
 - Sodium stearate, sodium alginate and methyl cellulose
- Conditioning agents: used to improve the the manageabilty and texture of the hair
 - Lanolin and its derivative
- Preservatives

Hair cosmetics...

Ideal Properties of a shampoo

- Should be non-toxic, non-irritant and should not damage the skin and sensitive organs like eyes if accidentally dropped to the eye
- Must have an appropriate cleansing ability
- Must produce adequate foam in soft and hard water
- Must protect the hair from friction damage during washing
- Hair must have a natural luster and an appropriate softness

Hair cosmetics...

Hair conditioners

- the primary purpose of a hair conditioner is to reduce the magnitude of the forces associated with combing or brushing of the hair, especially when wet
 - adds moisture and nourishment to hair strands and makes them soften, look soft, smooth, silky and shiny
 - accomplished by the deposition of conditioning agents that lubricate the hair fiber, diminishing surface friction and, therefore, combing forces
 - deposition of a conditioning agent also causes the hair to feel softer and more moisturized

- Another secondary benefit is the reduction or prevention of flyaway hair, especially by cationic conditioners
 - Besides making the hair more manageable, increasing the ease of combing also improves the ability to align the hair fibers in a more parallel configuration, which can result in an increase in hair shine, even if the shine of the individual fibers is not increased.
- Some ingredients can also form a film on the hair surface that provides color retention benefits for color-treated hair

Hair cosmetics...

- The commercial hair conditioners are found in different forms
 - thick Vaseline pomades, creams, gels, mousses, lotions, and spray mists
- Categorizing by application method
 - regular rinse off conditioners, intensive treatment conditioners, and leave-in products.

Regular rinse-off conditioners

- are normally applied after shampoo, followed by a rinsing step
- the most common form of conditioner sold

intensive treatment conditioners

- are used similarly to the above products, but are not meant for daily application

Hair cosmetics...

- Used for intensive treatment and a higher degree of conditioning
- contain a higher level of active ingredients that are kept on the hair for a longer period of time prior to rinsing
- marketed as thicker creams to provide higher conditioning

Leave-in products

- usually are lighter and can potentially provide more significant benefits than the above rinse-off products since everything applied stays on the hair until the next shampoo

Hair cosmetics...

- Found various forms, such as leave-in lotions, and sprays
- marketed either for single application or multiple applications during the day
- despite the different forms and positioning, most commercial conditioners contain the same general classes of conditioning agents with differences mainly in
 - Concentrations
 - numbers of different agents, and
 - particular members of a conditioning class employed.

Hair cosmetics...

The major classes of conditioning agents

Cationic Surfactants

- Cationic surfactants in the form of quaternary ammonium compounds (quats) are the most widely used conditioning agents
- include stearalkonium chloride, cetrimonium chloride, and dicetyldimonium chloride
- Because of the positive charge on the quats, they are substantive to hair and bind to negative sites on the hair surface.

Hair cosmetics...

- Treatment with these quats results in hydrophobic coating on the fiber that render the hair softer and easier to comb
- Buildup of static charge (flyaway) is also greatly reduced as a result of this surface modification
- Many important properties of quaternary ammonium conditioners are related to the degree of hydrophobicity of the lipophilic portion of the surfactant.

Hair cosmetics...

- Increasing the length of the alkyl chain of a monoalkyl quat and making it more hydrophobic leads to increased deposition on hair.
 - Cetrimonium chloride, as a result, deposits on hair to a greater extent than laurtrimonium chloride
- Increasing the number of alkyl chains also increases the deposition
 - tricetylmonium chloride exhibits greater deposition than dicetyldimonium chloride, which, in turn, is more substantive than the monocetyl quat.

Hair cosmetics...

Cationic Polymers

➤ There are numerous cationic polymers that provide conditioning benefits, especially improved wet combing and reduced static charge

○polyquaternium-10, a quaternized hydroxyethylcellulose polymer; polyquaternium-7, a copolymer of diallyldimethylammonium chloride and acrylamide; polyquaternium-11, a copolymer of vinylpyrrolidone and dimethylaminoethyl methacrylate quaternized with dimethyl sulfate; polyquaternium-16, a copolymer of vinylpyrrolidone and quaternized vinylimidazole; and polyquaternium 6, a homopolymer of diallyldimethylammonium chloride.

Hair cosmetics...

- By virtue of their cationic nature, the above polymers are substantive to hair
- The particular conditioning effectiveness of any these materials depends on the polymer structure
- Unlike common monofunctional quats, the first four of the above polymers are compatible to varying degrees with anionic surfactants
 - are used more often in shampoos than in stand-alone conditioners, although they find some use in leave-in conditioners
 - Polyquaternium-10 (PQ-10) and polyquaternium-7 (PQ-7) are most frequently used polymers in commercial shampoos.

Hair cosmetics...

Silicones

- used as conditioning agents in a wide variety of products including conditioners, shampoos, hair sprays and gels
- One of the most widely used silicones is dimethicone (polydimethylsiloxane)
- Other important silicones are dimethiconol, dimethicone copolyol, amodimethicone

Shampoo Vs Conditioner

SHAMPOO	CONDITIONER
A shampoo is a soapy liquid used for washing hair	A conditioner is a liquid you can apply on your hair after you have washed it with shampoo
We first cleanse the hair with shampoo	Then we apply conditioner on the hair to moisturize it
Remove excess dirt, oil, and unwanted build-ups from the hair and scalp and clean them thoroughly	Add moisture and nourishment to hair strands, making the hair look soft, smooth, silky and shiny
Produce lather	Don't produce lather
Contain surfactants like ammonium laureth sulfate, ammonium lauryl sulfate, and Cocamidopropyl	Contain hydrating and moisturizing agents Dimethicone and Dimethiconol
	Visit www.PEDIAA.com

Hair-coloring dyes (tint)

- To give a younger appearance by making gray hairs less prominent
- To make hair more stylish by actively varying the color of the hair
- Classified on the basis of their degree of durability in to
 - Temporary hair colors, semi-permanent hair colors and permanent hair colors
 - The mechanism by which they color the hair is different from each other
- include dye containing creams, shampoos or conditioners prepared by using substance obtained from plants, metallic compounds and a mixture of these two types of substances.
- most commonly used organic material from plant origin is 'Henna'

HAIR COLOUR TYPE**AMOUNT OF TIME
COLOUR LASTS****LIFT PERFORMANCE /
GREY COVERAGE**

PERMANENT



Until hair grows out



Up to 3 shades lighter
Up to 100% grey coverage

NON-PERMANENT
OR DEMI-PERMANENT



Up to 24 washes



Very little lift
Blends away greys

NON-PERMANENT
OR SEMI-PERMANENT

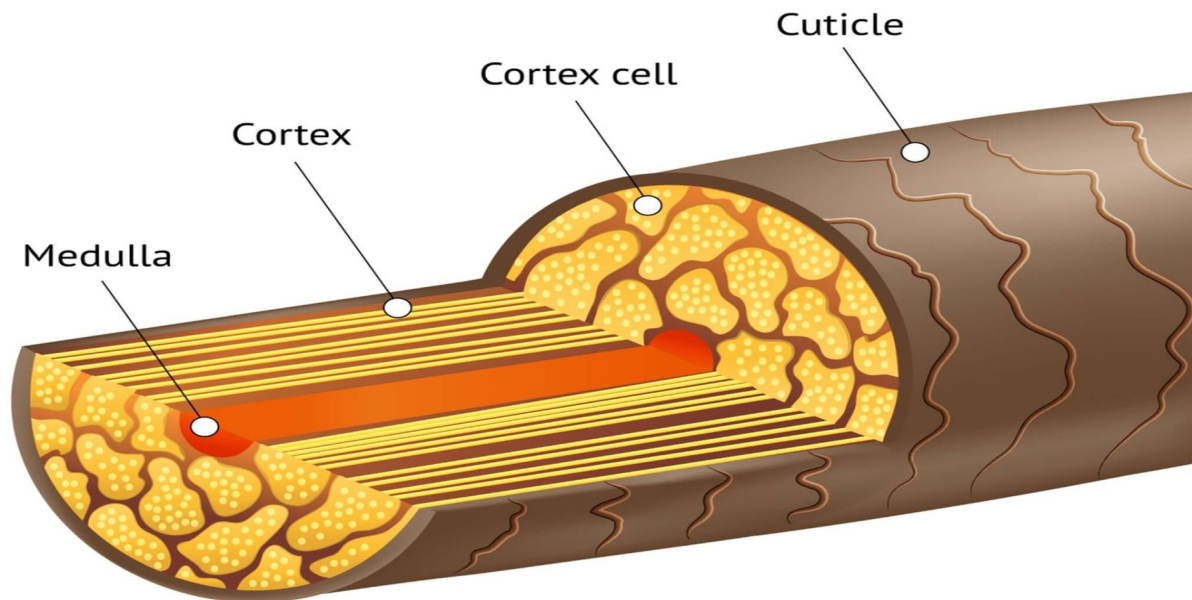


Up to 8 washes



No lift
Blends away greys

HAIR STRUCTURE



Hair cosmetics...

Characteristics of a hair dye

- Should be non-irritating to the hair and skin
- It should be non-toxic
- It should be able to color the hair shaft without damaging it
- It should be stable and should not change its colour when exposed to air, sunlight, water, shampoo, or hair conditioning agent

Hair cosmetics...

Hair Tonics

- Are preparation which are applied to the hair for curing baldness, relieving oily or dry skin and to prevent or cure dandruff
- Contains a combination of at least 2 to 3 ingredients i.e sebaceous gland stimulant (oily or waxy matter, called sebum), rubefacient (Produces Redness) and an antiseptics

Hair cosmetics...

Hair growth promotor

- Are preparations made by adding various pharmaceutical agents to an alcohol-water solution
- Is applied to the scalp to normalize its functions
- Help to prevent dandruff and itchiness
- Depending on the type and amount of active ingredient in them and their efficacy, classified as cosmetics, quasi drug (products that fall in between cosmetics and pharmaceuticals), OTC product and ethical pharmaceutical products

Hair cosmetics...

- **Minoxidil** promotes a growth of fine hair any place on the skin where there are hair follicles.
- sold under the trade name **Rogaine**



Before Minox



After Minox



Hair cosmetics...

Hair removers (depilatory)

- Used to remove unwanted hair from the body.
- When the hair is removed by chemical method it is known as **depilation**.
- When the hair is removed by plucking, electrolysis or x-ray it is known as **Epilation**
- Are available as solid, semisolids (as cream form)



Hair removal

Hair cosmetics...

➤ The chemicals used for preparation of depilatories

- Metallic sulphide of barium, calcium and strontium, calcium thioglycerol and calcium thioglycollate

Good depilatory should

- complete its action within 2-5 minutes and during this period it should make the hair soft, swelled and dispersed which should be removed from the skin wiping or rinsing
- be non-toxic and non-irritant to the skin
- be easy to apply
- be economical to use
- elegant, odorless or pleasantly flavoured
- remain stable in the container

Face cosmetics

Face powder

- is a cosmetic powder applied to the face to set foundation
- can also be reapplied throughout the day to minimize shininess caused by oily skin
- The ingredients used in formulation of face Powder includes talc, kaoline, calcium carbonate magnesium carbonate, zinc stearate, magnesium stearate, calcium carbonate, color and perfume.



Face cosmetics...

There are different forms in which face powders are available

- **Loose powder:** the powder is the main component
- **Compact powder:** a solid form in which a small amount of oil has been added as a binding agent
- **Paper powder:** powder is applied to paper
- **Liquid face powder:** powders are dispersed through an aqueous solution
- **Kneaded powder:** has glycerin or some other solution kneaded in to it

Face cosmetics...

Characteristics of face powder

- It should be very fine and should not have any gritty particles
- The ingredients should be evenly distributed
- It should be non-toxic and non-irritant to the skin
- It should be chemically and physically stable
- It should spread easily and adhere to the skin
- It should have good absorbing property
- It should remove shine from the face
- It should cover minor imperfections of the face and should not appear as paint
- It should look natural
- It should not dust off in a few minutes and make repowdering necessary

Face cosmetics...

A concealer or colour corrector

- is a type of makeup used to cover pimples, dark circles and other small blemishes visible on the skin.
- mainly used by women but also by some men



Rouge (Blush)

- Is a cosmetic typically used by women to redden the cheeks to provide a more youthful appearance, and to emphasize the cheekbones
- The form of rouges are compact, liquid, cream and stick



Face cosmetics...

CONCEALER	CORRECTOR
Concealer is a cosmetic product that helps to cover up imperfections on your skin	Corrector is a product that uses complementary colours to cancel out skin discolouration
Flesh-coloured and are supposed to match your skin tone	Have different colours, mostly pink, salmon, orange, purple, yellow and green
Conceal any flaw	Neutralize the effect of the flaw
Used after the corrector	Used before the concealer
	Visit www.PEDIAA.com

Face cosmetics

The quality requirements for rouges

- They should fit in well with foundations and it should be easy to smooth off the brush marks
- There should be no change in color
- They should have a suitable coverage, luster and adhesion
- They should be easy to wipe off and should not stain the skin

Face cosmetics...

Foundation

- skin colored cosmetics which give uniform color the [face](#) and conceal flaws (defect) to produce an impression of health and youth.
- Usually a liquid, cream, powder.
- Found powder compact form, oil-based foundations and emulsion foundations (O/W or W/O)



Face cosmetics...

Cold cream

- Cosmetic preparation which applied to smooth face and remove makeup
- derives from the cooling feeling that the cream leaves on the skin
- Prepared by emulsification oils and water (W/O)

E.g Beeswax-borax cream

Vanishing cream

- Are O/W emulsions which are prepared by emulsification of stearic acid and water
- seem to disappear when applied on skin

Eye makeups

Eye liner

- is a cosmetic used to define the eyes.
- applied around the contours of the eye
- To create a variety of aesthetic appearance.



Eye shadow

- Is applied to eyelids and corners of the eyes to create shadow and produce a sense of relief in order to emphasize the beauty of the eyes
- Has a greater variety of color than any other eye makeup



Eye make ups...

- The color ranges from the normal ones of blues, violet, browns and grays to the more vivid ones like greens, oranges and pinks.
- Colored titanium dioxide coated micas (pearly pigments) are used for coloring

Eye make ups...

There are different types of eye shadows

Liquid-paste form

- Oil-based and Emulsion type (W/O type, O/W type)

Solid form

- Powder compact
- Oil-based stick form
- Pencil type
- ❖ The main type is the powder compact in which the eye shadow has been pressed in to a container

Eye make ups...

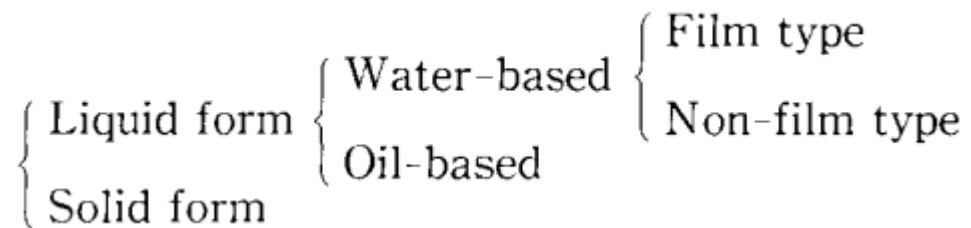
The following are the quality requirements for eye shadows

- They should be easy to apply smoothly and have good skin adhesion
- They should have no oily luster when applied
- There should be no change in color
- They should not be spoiled by sweat or sebum and maintain a good appearance
- They should be highly safe as they used around the eyes

Eye make ups...

Mascara

- is a cosmetic used to darken, thicken and define eyelashes.
- found in different forms



- The general purpose of mascara is to **emphasize, thicken, lengthen, and define lashes.**
- Mascara is available with **tube** and **wand** applicators.

Eye make ups...

Quality requirement for mascara are as follows. They should

- be non-irritating as they are applied to the eyes
- go on evenly and should not harden the eyelashes
- make the eyelashes look thick and long
- have an appropriate luster
- have an appropriate drying time
- not go on the lower eyelids and their appearance must not be spoiled by sweat, tears or rain
- be easy to remove
- should be easy to use through their period of use
- not be contaminated by microorganism

Eye make ups...

Eye wrinkle care products

- the skin around the eyes becomes dry easily than on any other part of the face.
- If the skin left in dry conditions this will cause wrinkles.
- This is prevented by eye wrinkle care products which have a humectant effect.
- They are different types
 - Oil-based (stick type) and emulsion types (creams or milky lotions)

Eye make ups...

Eyebrow cosmetics

- Are used after adjusting the form of the eyebrows with razors or scissors or by plucking out hairs to draw in the desired form, to make darker in color or brighter
- The most common are the easy-to-use pencil and mechanical pencil types but there are also powder compact and liquid types.

Eye makeup remover

- To remove the different types of makeup
- The main types are for removing water and oil-based makeup

Lip decorators and carers

Lip gloss

- is a cosmetic product used primarily to give the lips shine and sometimes subtle color.
- is distributed as a liquid or a soft solid.
- can be completely clear, translucent, or various shades of opacity



Lip balm

- is a lip cosmetic product which generally has medicinal purposes



Lip decorators and carers...

Lipstick (Lippy)

- is a cosmetic product containing pigments, oils, waxes, and emollients that applies color and texture to the lips.
- Consists mainly of an oily base material and coloring agents
- Oil base material contain
 - Waxes which are solid at ordinary T are used to provide the stick form of lipsticks
 - Natural waxes such as carnauba wax, beeswax candelilla wax



Lip decorators and carers...

- Oils used are those that are liquid at ordinary temperature or whose melting points are around body T
- Natural oils such as cocoa butter, castor oil, jojoba oil, lanoline oil; hydrocarbon oils such as petrolatum and liquid paraffin and synthetic fatty acid esters

Coloring agent

- The synthetic organic colors are used in lipsticks which could be Pigments or dyes
- Pigments determine the color of lipsticks
- To improve the lasting power of the color on the lips, staining dyes are used as well
- Eg tetrabromofluorescein, tetrachlorotetrabromofluorescein and dibromofluorescein

Lip decorators and carers

- A variety of inorganic pigments are used in addition to the synthetic organic colors
- Titanium dioxide, red iron oxide, yellow iron oxide and black iron oxide are used to adjust the tone and brightness
- Titanium dioxide coated mica and colored titanium dioxide coated mica use to give a pearly luster
- Other inorganic pigments are treated using various methods in order to enhance their dispersion and stability

Lip decorators and carers...

Quality of good lipstick. It should

- be non-toxic and non-irritating
- be free from gritty particles
- be easily applicable and easily removable if need arises
- be good odour
- give shine and smooth appearance
- be not dry on storage
- maintain its lip colour long after application
- not break during use and it should maintain its firmness till whole of it is used up

Lip decorators and carers...

- **Lip Plumper** is a cosmetics product used to give the lips an enhanced, fuller look.
- Lip Plumper is distributed as a viscous liquid applied to the lips with a roller or brush applicator.



Lip decorators and carers...

- It can be translucent or have various tints, and can be worn alone or over regular lipstick.
- work by irritating the thin, delicate skin of the lips with ingredients like menthol, camphor cinnamon or cayenne pepper.
 - This makes the lips swell slightly, which may also diminish the appearance of fine lines in that area
- A lip plumper's effects are temporary, so it must be reapplied throughout the day to maintain the results.

Nail cosmetics

Nail polish



- nail varnish or nail lacquers or nail enamels
- is a lacquer that is applied to the nails of both the fingers and toes, usually cosmetically,
- The act of simply polishing the nails without applying a chemical layer afterwards is called nail buffing



Nail cosmetics...

- nail polishes have an **aesthetic advantage**: they offer more refinement and elegance and allow for a wide range of personal preferences.
- the nail varnish plays **a protective role**: it forms a film on the nail surface, which is both tough and flexible, increasing the resistance of the nail to microtraumas.
- It also maintains a more constant degree of **nail hydration**.

Nail cosmetics...

- The main constituent of the film that remains on the nail after drying (evaporation) is nitrocellulose.
- The film former has a lot of qualities
 - hard, tough, stable, and waterproof,
 - but it is neither sufficiently adherent nor glossy or flexible.
- To improve adherence and gloss, film modifiers are added
 - The most common one is Santolite 1 or toluene sulfonamide/formaldehyde resin (TSFR).

Nail cosmetics...

- With the nitrocellulose and the film modifier, the resulting film is hard, tough, adherent, and glossy. Unfortunately, it is still not flexible enough and cracks.
- Plasticizers are added to increase flexibility.
 - These are molecules with a high boiling point, which remain in the film after drying.
- They reportedly increase separation between the cellulose links as well as increasing the rate of solvent evaporation
 - Eg. castor oil, glyceryl tribenzoate, acetyl tributyl citrate, PPG-dibenzoate, glycerol, citrate esters, triacetin, and a polymer plasticizer called “NEPLAST” (a polyether-urethane).

Nail cosmetics...

- the nail polish bottle may also have solvents and Thinners , pigments and dyes, thixotropic agents, as well as other additives
- The most commonly used solvents are alkyl esters (ethyl, N-butyl acetate) and glycol ethers (propylene glycol monomethyl ether).
- These solvents, which have different boiling points and evaporation rates, allow the regulation of drying time
- They also allow lower viscosity, which improves brush ability.

Nail cosmetics...

- Thinners are not real nitrocellulose solvents but are miscible with them.
 - Thinners help regulate the evaporation rates and stabilize viscosity.
 - a polish low in viscosity is easy to apply and leaves a homogeneous film.
 - If viscosity is too low, the coverage of the film will be insufficient.
 - If viscosity is too high, the film will be thick and streaky.
 - E.g. ethanol, isopropanol, butanol and toluene.
- Pigments and colorants also need to draw the attention.
- Some are soluble in nitrocellulose and originate transparent polishes, very slightly colored.

Nail cosmetics...

- Most of them are not soluble and originate the most-used polishes (the nail enamels)
 - Eg. Mineral pigments: ferric ferrocyanide (Prussian blue), titanium dioxide (TiO_2) and Organic pigments: a barium lake, calcium lakes and zirconium lake
- Thixotropic agents increase nail enamel viscosity at rest, thus preventing pigment precipitation.
- They, however, become fluid as soon as a mechanical constraint is exerted, either by shaking the brush or the bottle itself

Nail cosmetics...

The perfect nail polish

- has to be easy to apply, and it should dry quickly
- should leave a shiny, smooth, even, hard, and flexible film that is able to last five days
- should be removed without leaving any trace and should not have any side effects
- should be stable in the bottle
- should offer a wide range of colors, enabling one to get the expected aesthetic effect

Nail cosmetics...

Unwanted Effects of Nail Polishes

- The nail polishes can cause an orange staining of the nail plate
- Nail lacquers can cause keratin granulation
- Nail varnishes can also cause allergic contact dermatitis, more rarely, contact urticaria



Keratin granulation



orange staining of the nail plate

Nail cosmetics...

Nail Polish Remover

- dissolve nitrocellulose and remove lipids from the nail plate.
- They mainly contain a mixture of organic solvents, with small amounts of oils added to counteract the drying effect of the solvents.

Typical formula

Solvents (98%)—Example: acetone, butyl or ethyl acetate, ethoxyethanol.

Lipids (2%)—Example: castor oil, lanolin oil.

Nail cosmetics...

- Other additives can also be found such as dyes, fragrances, preservatives, vitamins, and UV absorbers.
- should not be used more than once a week and should not be left too long in contact with the nails and the skin

Skin cosmetics

➤ Cosmetics are applied to the dead cells of the corneal layer.

The basic function of skin cosmetics

- Anti-drying, cleansing, ultraviolet damage prevention, anti-oxidation and invigoration
- Clean up skin problems
- Whitening effect to combat skin aging associated troubles
- Prevent wrinkles, sagging skin and acne
- To mask body odor

Skin cosmetics...



Depilatories

- are sulfides of calcium and barium
- have the property of removing hair from the skin

Bronzers

- are cosmetic product that darken skin color
- they found in the form of creams, powders, sprays, gels, and lotions .



Skin cosmetics...

Antiperspirants and deodorants

- The major components of body odor are products decomposed from a mixture of a sweat and lipid broken down by the normal bacterial flora of the skin.
- One group of unpleasant smelling breakdown product is the lower FAs
- Cosmetics for preventing body odor contain antiperspirant perfumes and antibacterial agents
- The principal antibacterial agents are benzaalkonium chloride, benzethonium chloride, chlorhexidine chloride etc

Skin cosmetics...

Antiperspirants



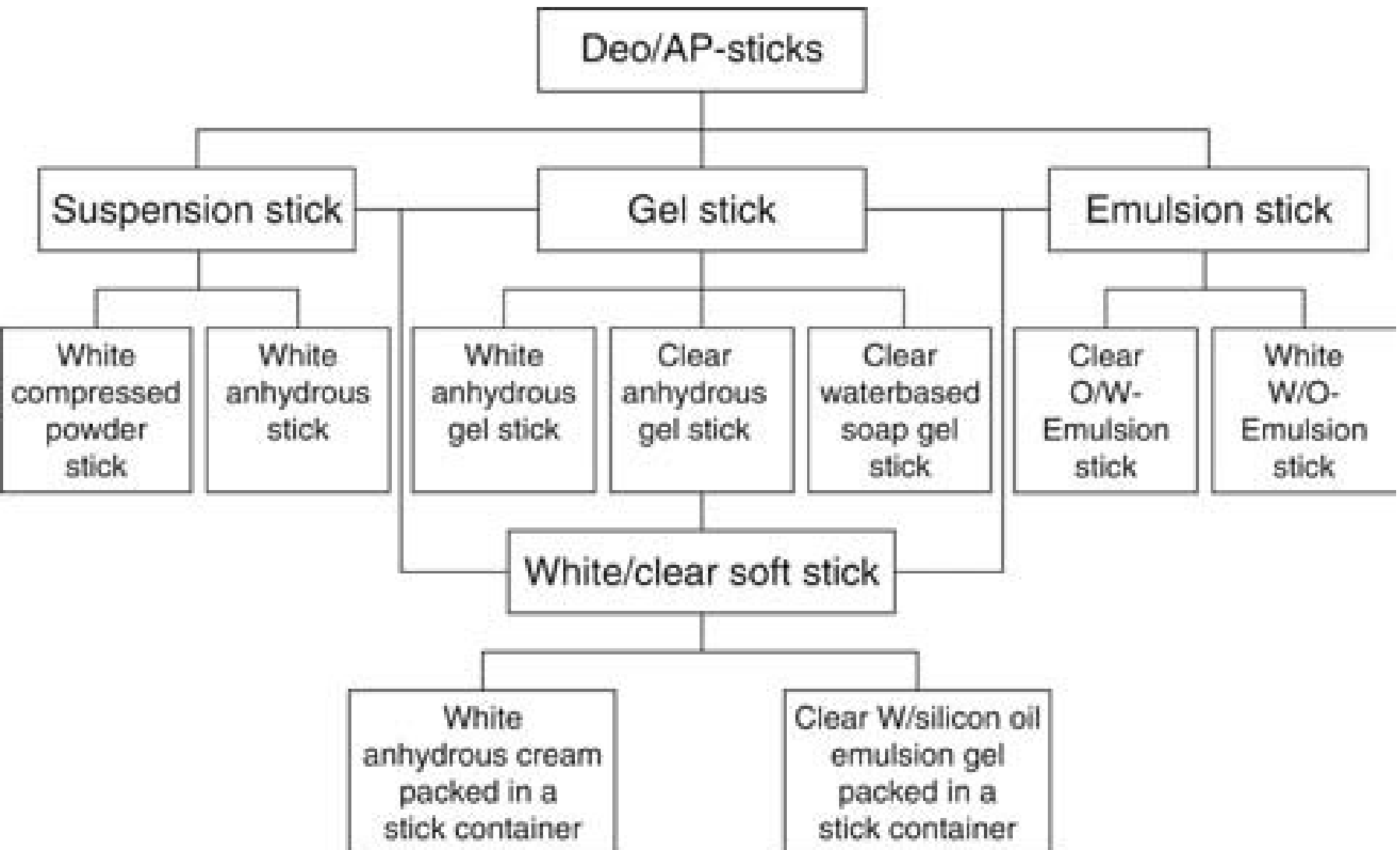
- are topically applied products designed to reduce underarm wetness by limiting eccrine sweat production
- Suppress production of perspiration due to their strong astringent action on the skin
- Active Ingredients for Antiperspirants:
 - Zirconium and aluminum chlorides, **aluminium hydrochloride**, potassium aluminum sulfate, zinc oxide and zinc paraphenol sulfonate
 - Aluminum hydrochloride is used most often

Skin cosmetics...

Application forms for antiperspirant actives

- Typical application forms for antiperspirants are sticks, roll-ons, creams, pump sprays, aerosols, gels, and powders
- The three most important product forms are sticks, roll-ons, and aerosols
- Sticks application form that is prepared in a relatively long and slender, often cylindrical, form
- Roll-on application form applied to the underarms with a rolling applicator.
 - has a "ball" that rolls the antiperspirant on the under arm skin.

Skin cosmetics...



Skin cosmetics...

Antiperspirant Sticks

- Sticks can be divided into different classes like suspension sticks, gel sticks, emulsion sticks and Soft sticks .
- Soft sticks have some properties of all the three categories

Suspension sticks

- the active in the form of a powder is suspended in a silicone oil phase
- Stearyl alcohol is used as the hardening agent
- Quaternium-18 hectorite can reduce the settling of the actives
- Cyclomethicones give the stick a dry, silky feel; nonvolatile oils, like PPG-14 butylether, minimize white residues on the skin

Skin cosmetics...

Gel Sticks

- Subdivided into the following groups: white anhydrous gel sticks, clear anhydrous gel sticks, and clear water-based soap gel sticks

White Anhydrous Gel Sticks :

- sticks with improved washout performance compared with the classic suspension sticks
- They contain N-acyl amino acid amides and 12-hydroxyacid as gelling agents
- The washout agent is an ethoxylated solubilizer, like Cetareth-20.
- These white sticks turn clear after application to the skin (no-residue stick).

Skin cosmetics...

Clear Anhydrous Gel Sticks

- No white residue on skin, no dangerous ingredients, and high efficacy
- A typical gelling agent is dibenzylidene sorbitol (dibenzylaldehyde monosorbitol acetal)
- The sticks usually contain a high level of alcohol and/or polyols.
- At high polyol concentration, the active is regarded to be solubilized instead of suspended in the gel matrix

Skin cosmetics...

Emulsion Sticks

- grouped into clear O/W emulsions and white W/O emulsions
- **Clear O/W Emulsions** : contain a high surfactant combination with the active solubilized in the external water phase.
 - The high concentration of surfactants is a disadvantage
- **White W/O Emulsion Sticks** : the active is solubilized in internal water by a surfactant, like polyglycerol-4 isostearate.

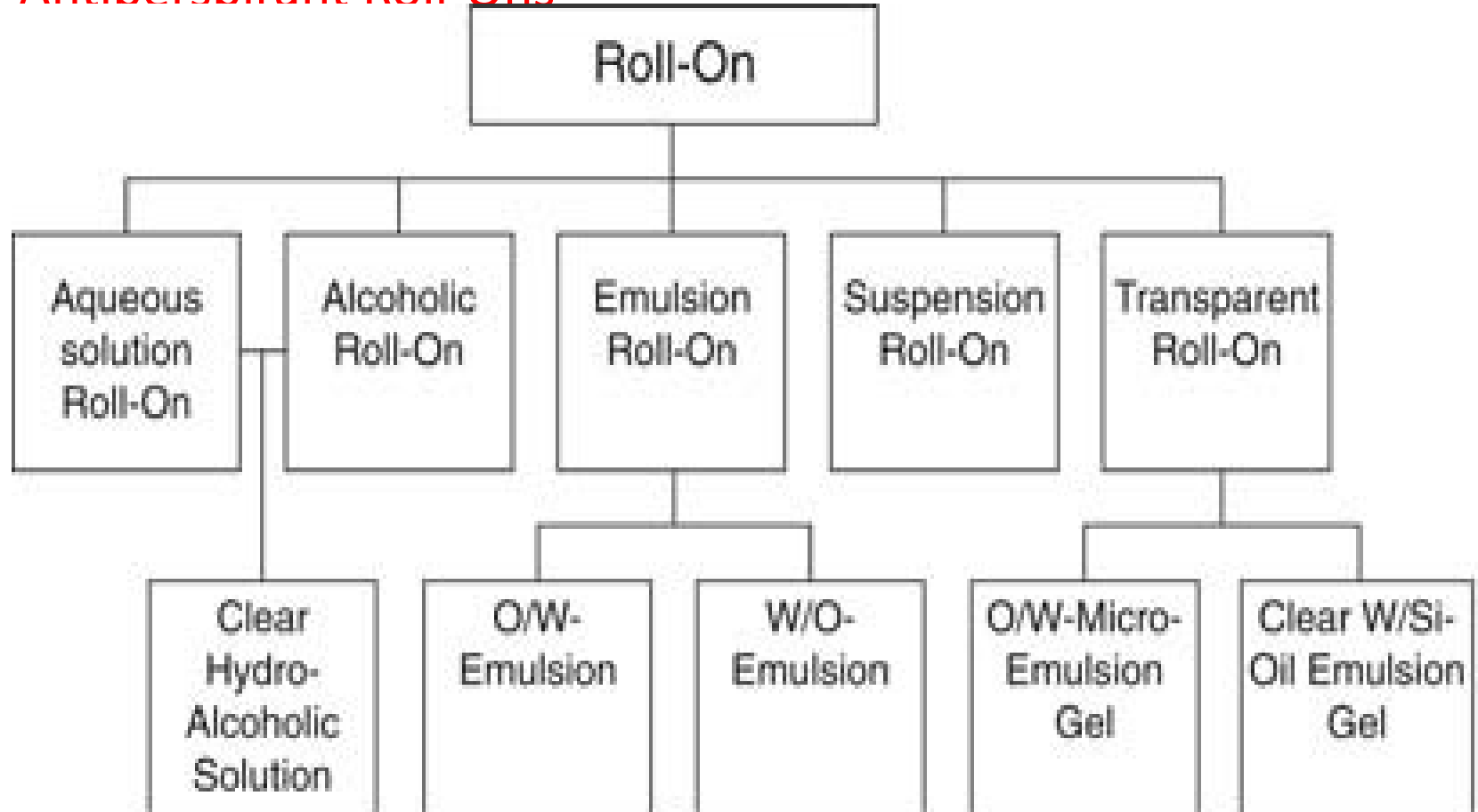
Skin cosmetics...

Soft Sticks (Soft Solids)

- Differentiated into two subgroups:
 - **White, Anhydrous suspensions:** contain an antiperspirant active, a volatile and nonvolatile silicone oil, and a thickener (N-acyl glutamic acid amide).
 - **Clear Water-in-Silicone Emulsion Gels:** achieved by adjusting the refractive index of the water and silicone oil phase.
 - Silicone formulation aids are mixtures of cyclomethicone and dimethicone copolyol helping to solubilize the active
 - Low surface tension of cyclomethicones facilitates good spreading of a product on the skin and reduces the tackiness of antiperspirant actives

Skin cosmetics...

Antiperspirant Roll-Ons



- Have nongreasy and nonoily feel in the axilla and good spreadability of the content on the underarm skin

Skin cosmetics...

Clear Hydroalcoholic Roll-On

- contains a water/alcohol solution of the antiperspirant active thickened with a water-soluble polymer like hydroxyethylcellulose.
- The alcohol in the formula gives a fresh sensation in the axilla and facilitates drying of the product.
- Excellent antiperspirant efficacy is another benefit of hydroalcoholic roll-ons.

Skin cosmetics...

O/W Emulsion Roll-On

- The active is dissolved in the outer phase, allowing the formulation of a highly effective product.
- In alcohol-free formulated systems, microbiological stability has to be checked

W/O Emulsion Roll-On

- weaker in efficacy because the actives are encapsulated and the external oil phase often gives a sticky feeling

Skin cosmetics...

Transparent roll-on

- Includes W/Si Emulsion Roll-On and O/W Micro-Emulsion Gel

W/Si Emulsion Roll-On

- Silicone oils allow products to formulate on the basis of a “W/O technology,” because the skin feel is not comparable with traditional oily components, like ester oils or triglycerides

O/W Micro-Emulsion Gel

- An alternative approach to transparent products uses the phase inversion temperature (PIT) technology.

Skin cosmetics...

- A suitable mixture of surfactants, oils, and water is heated from 60.8 °C to 90.8 °C to give a W/O emulsion above the PIT.
- During cooling, the mixture shows phase inversion to give white or transparent O/W emulsions; O/W micro-emulsion gels are obtained in the presence of hydrophobically modified water-soluble polymers.

Skin cosmetics...

Suspension Roll-On

- The antiperspirant active in powder form is suspended in cyclomethicone.
- The roll-on can be formulated with or without ethanol.
- Quaternium-18 hectorite is used as a thickener to prevent settling of the active.
- it does not give a wet feeling after application because of the easy drying

Skin cosmetics...

Antiperspirant Aerosols

- Typical ingredients for aerosols include isopropyl myristate, isopropyl palmitate, volatile silicone, dimethicone, silica, clays, propylene carbonate, and ethanol
- Propellants include propane, butane, and isobutane
- formulated as water-free suspensions
 - The active is suspended as a powder in an oil phase like cyclomethicone
- Agglomeration of solid particles and settling of actives can be minimized by the usage of suspending agents like fumed silica (amorphous silicon dioxide) or clays (bentonite and hectorite).

Skin cosmetics...

- The clays form a weak gel in the presence of an oil phase that can be destroyed by shaking the aerosol can before usage.
- The gel structure is reformed on standing, thereby holding the active in suspension.

Skin cosmetics...

Deodorants

- are topically applied products designed to reduce underarm odor.
- They are considered in the United States as cosmetics, while antiperspirants are treated by the FDA as drugs
- Deodorants tend to be less irritating than antiperspirants

Skin cosmetics...

There are different types deodorant with different strategies to reduce underarm

- Antiperspirant active-containing deodorants
- Odor-masking deodorants
- Odor-neutralizing deodorants
- Odor-quenching deodorants
- Enterase inhibitors
- Antimicrobial active-containing deodorants

Skin cosmetics...

Antiperspirant active-Containing Deodorants

- Antiperspirant actives such as aluminum chlorohydrate or the Al–Zr complexes reduce the secretion of eccrine sweat

Odor-masking deodorants

- Fragrance compositions (such as perfumes) are used to mask odors
- It is conventional to incorporate 0.2% to 1.5% of a perfume in body deodorants
- They are designed to blend with the underarm odor and thus act as a masking agent

Skin cosmetics...

Odor-Neutralizing Deodorants

- Odorous lower fatty acids contribute to underarm odor.
- Chemical neutralization with sodium bicarbonate (NaHCO_3) yields the corresponding odorless soaps
- This active, however, is not stable for a long time in aqueous compositions

Skin cosmetics...

Odor-Quenching Deodorants

- **Zinc Ricinoleate** Zinc salts of ricinoleic acid have no bacteriostatic or antiperspirant effect
- They strongly bind odorous fatty acids, amines, and mercaptanes
- Ligand-exchange reactions of ricinoleic acid for odor molecules are probably the reason for the quenching properties of zinc ricinoleate
- **Metal Oxides**: The oxides of calcium, magnesium, and zinc in the presence of fatty acids form the corresponding metal soaps

Skin cosmetics...

Esterase Inhibitors containing deodorants

- **Zinc Glycinate**: inhibit exoenzymes from the underarm bacteria and reduce under arm odor.
- **Triethylcitrate** : The optimal pH value for development of underarm odor caused by coryneform bacteria is approximately about pH 6 in axillary extracts.
 - Shifting the skin surface pH to the acidic side decrease the activity of skin esterases, which are proposed to be responsible for degradation of underarm secretions.
 - Triethylcitrate was proposed to form citric acid by an enzymatic process on the underarm skin.

Skin cosmetics...

Antimicrobial Active-Containing Deodorants

- ❖ Triclosan (2,4,4'-Trichloro-2'-Hydroxydiphenylether):
 - This active has a broad-spectrum antimicrobial activity against most gram-positive and gram negative bacteria, molds, and yeasts.
- ❖ Glyceryl Fatty Acid Ester: Mono- and oligoglyceryl fatty acid esters such as glyceryl monocaprylate, monocaprylate, monolaurate, and diglyceryl monocaprylate
 - higher biodegradability and selective bacterial action

Skin cosmetics...

- Combinations of glyceryl monolaurate with farnesol and phenoxyethanol showed synergistic efficacy effects against coryneform bacteria

❖ Sucrose Fatty Acid Ester:

- Sucrose substituted with fatty acids on hydroxyl groups
 - Eg. Sucrose monostearate and sucrose monolaurate
- The antimicrobial potential depends strongly on the substitution degree of the sucrose

Skin cosmetics...

- ❖ **Glyceroether:** 2-Ethylhexyl glyceroether (octoxyglycerol) is a clear liquid with good solubility in cosmetic oils, polyols, and alcohol but only moderate solubility in water (0.2%).
- Synergistic antimicrobial activity with other ingredients

Skin cosmetics...

Application forms for deodorant actives

- Typical application forms are sticks, roll-ons, creams, pump sprays, aerosols, and gels
- Sticks, pump, and aerosols most common
- Deodorants can be formulated in acidic, neutral, or alkaline environment.

Skin cosmetics...

Deodorant Sticks

- are solidified by 6% to 8% of sodium stearate.
- The deodorizing agent and a fragrance are dissolved in a hydrophilic carrier.
- Two types: the **ethanol-based** and the **propylene glycol-based** sticks.
- Ethanol-based sticks are preferred if it is the intent of the formulator to create a cooling sensation for the consumer.
- Shrinkage of the stick has to be taken into account because of evaporation of the alcohol
- Propylene glycol-based sticks tend to be more resistant to shrinkage, and solubilization of a fragrance is easier in some instances

Skin cosmetics...

Deodorant Aerosols

- Spray products containing a solution of an antimicrobial active in an ethanol and/or propylene glycol carrier, blended with a liquefied propellant
- The difference from an antiperspirant active containing aerosol is that the deodorizer is solubilized in an alcohol- or polyol-based formulation and not suspended.
- Deodorant sprays provide a dry skin feeling to the underarm skin because they are anhydrously formulated.

Skin cosmetics...

- 20% to 60% of the sprayable contents of an aerosol reach the skin, because the liquefied hydrocarbon propellant vaporizes as it is sprayed.
- Propane, butane, and isobutane are the most commonly used propellants

Skin cosmetics...

Deodorant Pump Sprays

- Pump sprays allow a good dosage of the formulation to be delivered to the underarm skin in a hygienic way
 - ❖ **Hydroalcoholic Pump Sprays:** consist of low viscosity hydroalcoholic solutions of a deodorizer and a perfume
 - a solubilizer is incorporated into the formulation to maintain a clear and homogeneous solution
- E.g PEG-40 hydrogenated castor oil

Skin cosmetics...

- **PIT-Emulsion Pump Sprays:** The sprayable low-viscous deodorant based on the phase inversion temperature (PIT) technology.
- Suitable mixtures of ethoxylated surfactants, oils, and water in the presence of antiperspirant and deodorizing actives are heated to 60 °C to 90 °C.
- Cooling the resulting W/O emulsion to room temperature yields, via a PIT process, a finely dispersed bluish-white O/W emulsion.
- contained a skin-friendly deodorizing combination of glyceryl monocaprylate and wool wax acids in an alcohol-free delivery system

Skin cosmetics...

Microemulsion Pump Sprays

- Sprayable alcohol-free and transparent pump sprays
- This O/W microemulsion can be obtained with and without the PIT technology but needs careful selection of ingredients and considerable fine-tuning.
- The main advantage compared with classical microemulsions is the low surfactant concentration (<10%).

Skin cosmetics...



Sunscreen formulation

- UV rays in sunlight triggers production of the pigment melanin, which then protects the deeper layers of the skin from damage.
- Excessive exposure to ultraviolet radiation causes premature aging of the skin and leads to skin cancer.
- Most sunscreen formulations block UV-B radiation while letting through the less energetic long-wave UV-A rays that promote production of melanin.
- creams and lotions



Skin cosmetics...

- UV filters used in cosmetic sunscreen formulations are roughly considered belonging to two groups
 - organic molecules deliberately selected for their UVR-absorbing capacities (i.e., **organic UV filters**)
 - particles that reflect, or scatter UVR. Particles may be inorganic (i.e., metal oxides) or organic (**microfine polymeric molecules**).
- PABA derivatives OMC (Octyl methoxycinnamate) is now used as a UV-B filter in about 90% of all sunscreen lotions.

Skin cosmetics...

- Contain chemicals that interfere with secondary reactions such as
 - generation of free radicals and reactive oxygen species (ROS) in the skin (antioxidant)
 - generation of inflammatory mediators (ant inflammatory)
 - exert long-term protective effects on degenerative skin damage

Oral cosmetics

- Divided into oral cleansing products and mouth fresheners
- oral cleansing products
- Preparation for external use which clean the teeth and surrounding them, cleanse and refresh the inside of the mouth, prevent dental caries and periodontal disease
- Divided in to dentifrices and mouthwash

Oral cosmetics...

Dentifrices

- Preparation meant to be applied to the teeth
- are applied with a tooth brush for cleansing the accessible surfaces of the teeth
- Enhance appearance of the teeth, reduce tooth decay and bad odor of the mouth and make the gum healthy
- Dentifrices are found in the form of tooth powder, toothpaste and liquid-form dentifrice
- Ingredients include: polishing agents (abrasive), surface active agents (detergent), humectants, binders, sweetening agents, Flavoring agents, preservatives

Ingredients of Dentifrices

Ingredient category	Ingredients	Effect • Action
Abrasive	• Calcium carbonate • Calcium phosphate, dibasic • Silica • Aluminium hydroxide, etc.	• Principal ingredient of dentifrice • Remove dirt without scratching the tooth surface
Humectant	• Glycerin • Sorbitol, etc.	• Give moistness and plasticity (cream form) to the powders in the dentifrice
Foaming agent	• Sodium lauryl sulfate, etc.	• Disperse the dentifrice throughout the mouth and clean away the dirt in the mouth
Binder	• Sodium carboxymethyl cellulose • Carrageenan, etc.	• Bind the powder and liquid ingredients together and give moldability
Flavoring agent	• Sodium saccharin • Peppermint oil • Spearmint oil • Menthol, etc.	• Adjust flavor • Make dentifrice more palatable by giving a refreshing taste and smell
Coloring agent	• Permitted colors	• Make the dentifrice more attractive looking
Preservative	• Ethyl paraoxybenzoic acid • Butyl paraoxybenzoic acid, etc.	• Prevent deterioration of dentifrice
	• Purified water	• Adjust viscosity and consistency of dentifrice
Pharmaceutical agent	• Fluoride • Antibacterial agents • Anti-inflammatory agents, etc.	• Prevent dental caries, pyorrhea and gingivitis, stop halitosis and enhance effectiveness in removing oral cavity contaminants

Oral cosmetics...

Qualities of dentifrices

- It must remove foreign particles, food substances, plaque and clean the teeth
- Must be non-toxic
- Must be properly flavored and sweetened
- Must leave the mouth with a refreshing after-taste

Oral cosmetics...

Mouth wash (Mouth rinse)

- Similar in form to a liquid-form dentifrice but used without toothbrush
- An appropriate amount is put in the mouth to rinse it out and the discarded afterwards
- Mouthwashes are classified by usage and form in to
 - **Directly used type**, concentrated type and powder type

Oral cosmetics...

- Use to clean the inside of the mouth, prevent halitosis and refresh the mouth
- Ingredients: ethanol, and other solvents, humectants, solubilizers, flavoring agents, preservatives and pH regulators
- In powder type, sodium bicarbonate and other diluting agents are used in the place of liquid ingredients.

Ingredients of Mouth wash (Mouth rinse)

Category	Ingredient	Effect • Action
Water	Purified water	Adjustment of viscosity, consistency, volume, etc.
Solvent	Ethanol, etc.	Dissolution of flavoring agents, give refreshing feeling
Humectant	Glycerin, etc.	Moisturizing of mouth, aid dissolution of flavoring agents
Solubilizer	Polyoxyethylene-hydrogenated castor oil, polyoxyethylene-polyoxypropylene glycol, sodium lauryl sulfate, etc.	Solubilization of flavoring agents, cleaning inside of mouth
Flavoring agent	Sodium saccharin, Menthol, Peppermint oil, etc.	Give a cool, refreshing feeling, special flavor
Preservative	Ethyl paraoxybenzoate, Sodium benzoate, etc.	Prevent product deterioration
Coloring agent	Permitted colors, caramel, etc.	Color to give attractive appearance
pH adjuster	Phosphoric acid salts, citric acid salts	pH adjustment
Pharmaceutical agent		Stop halitosis, prevent dental caries, periodontal diseases, etc.

Oral cosmetics...

Mouth Freshener

- Is preparation for internal use designed to prevent nausea and other feelings of discomfort
- They found in the form of pills, lozenge and liquids
- The liquid form is packed in a squeeze or pump type container and is squirted in to the mouth as a mist
- Ingredients used to flavor mouth fresheners have a strong freshening effect and include
 - cinnamon , mint oil, fennel oil, D-camphor, L-menthol
 - Depending on the form of mouth refreshner, other diluting agents, flavoring agents and preservatives are also used

assignment

1. Write short note on
 - a) similarity and difference b/n drugs and cosmetics.
 - b) cosmetics safety
2. Write detailed explanation for
 - a) “cosmeceuticals”
 - b) quaisi drugs.



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